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Reranking partisan animosity in algorithmic social media feeds alters affective polarization

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Full article and list of author affiliations: https://doi.org/10.1126/science.adu5584

INTRODUCTION: Social media algorithms profoundly impact our lives, yet our understanding of these algorithms' impact has been limited to interventions that the platforms themselves are willing to test and publish. In this article, we demonstrate a method for independent algorithmic reranking experiments in naturalistic settings and at scale. We leverage this method to demonstrate that algorithmic ranking can both raise and lower levels of affective political polarization in the United States, with no detectable differences across party lines.

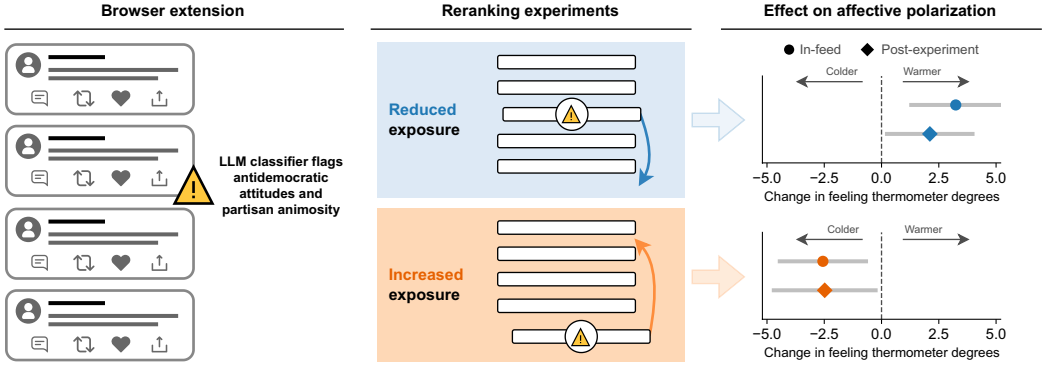
RATIONALE: We built a browser extension that intercepted and reranked web-based social media feeds in real time. The extension used a large language model (LLM) to score content and then applied these scores to reorder the feed, all without requiring platform cooperation. Our approach identified content expressing antidemocratic attitudes and partisan animosity (AAPA) and controlled users' exposure to such content through reranking.

We conducted a preregistered field experiment with 1256 participants on X, the most widely used social media platform for political discourse in the United States, during the weeks leading up to the 2024 presidential election. Participants were randomly assigned to two parallel experiments in which their feeds were dynamically reranked for 1 week to either up- or down-rank AAPA posts. We measured the effects of these interventions on affective polarization (participants' feelings toward the political outgroup) and emotional experience (anger, sadness, excitement, or calm) by using both in-feed and post-experiment surveys. Results were compared to those of participants in control conditions whose feeds were not reranked.

RESULTS: Reranking to change levels of exposure to AAPA content significantly influenced affective polarization. Increased AAPA exposure led to colder feelings toward the political outgroup, whereas decreased AAPA exposure led to warmer feelings. Both shifts amounted to more than 2 degrees on a 0- to 100- degree "feeling thermometer." The impact on affective polarization was consistently detected in both in-feed and post-experiment surveys, with no evidence of partisan asymmetry. Changes to the feed algorithm caused in-feed but not post-experiment changes to participants' negative emotions.

CONCLUSION: This work demonstrates a new method for reranking social media feeds that enables testing interventions without platform collaboration. We found that increasing exposure to AAPA posts significantly increases affective polarization and negative emotions, whereas decreasing exposure reduces them. These changes were comparable in size to 3 years of change in United States affective polarization. As political polarization and societal division become increasingly linked to social media activity, our findings provide a potential pathway for platforms to address these challenges through algorithmic interventions. Together, these interventions may result in algorithms that not only reduce partisan animosity but also promote greater social trust and healthier democratic discourse across party lines. □

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†These authors contributed equally to this work. **Cite this article as T. Piccardi *et al.*, Science 390, eadu5584 (2025). DOI: 10.1126/science.adu5584**



A platform-independent field experiment demonstrates that reranking content expressing antidemocratic attitudes and partisan animosity in social media feeds alters affective polarization. A browser extension reranked participants' social media feeds by reducing or increasing exposure to posts expressing AAPA, which in turn led to corresponding decreases or increases in affective polarization.

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SOCIAL MEDIA

Reranking partisan animosity in algorithmic social media feeds alters affective polarization

Tiziano Piccardi[1]*†‡, Martin Saveski[2]*†, Chenyan Jia[3]†, Jeffrey Hancock[4], Jeanne L. Tsai[5], Michael S. Bernstein[1]

Today, social media platforms hold the sole power to study the effects of feed-ranking algorithms. We developed a platform-independent method that reranks participants' feeds in real time and used this method to conduct a preregistered 10-day field experiment with 1256 participants on X during the 2024 US presidential campaign. Our experiment used a large language model to rerank posts that expressed antidemocratic attitudes and partisan animosity (AAPA). Decreasing or increasing AAPA exposure shifted out-party partisan animosity by more than 2 points on a 100-point feeling thermometer, with no detectable differences across party lines, providing causal evidence that exposure to AAPA content alters affective polarization. This work establishes a method to study feed algorithms without requiring platform cooperation, enabling independent evaluation of ranking interventions in naturalistic settings.

Social media algorithms profoundly impact our lives: They curate what we see (*1*) in ways that can shape our opinions (*2*–*4*), moods (*5*–*7*), and actions (*8*–*12*). Because of the power that these ranking algorithms have to direct our attention, the research literature has articulated many theories and results detailing the impact that ranking algorithms have on us (*13*–*17*). However, validating these theories and results has remained extremely difficult because the ranking algorithm behavior is determined by the social media platforms, and only the platforms themselves can test alternative feed designs and causally assess their impact. Platforms, however, face political and financial pressures that constrain the kinds of experiments they can launch and share (*18*). Concerns about lawsuits and the need to preserve engagement-driven revenue further limit what platforms are willing to test, leaving massive gaps in the design space of ranking algorithms that have been explored in naturalistic settings and at scale.

To address this gap, we present an approach that enables researchers to rerank participants' social media feeds in real time as they browse, without requiring platform permission or cooperation. We built a browser extension, a small add-on to a web browser that modifies how web pages appear or behave, similar to an ad blocker. Our extension intercepts and modifies X's web-based feed in real time and reranks the feed using large language model (LLM)–based rescoring, with only a negligible increase in page load time. This web extension allows us to rerank content according to experimentally controlled conditions. The design opens a new paradigm for algorithmic experimentation: It provides external researchers with a tool for conducting independent field experiments and evaluating the causal effects of algorithmic content curation on user attitudes and behaviors while preserving ecological validity.

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This capability allowed us to investigate a pressing question: Can feed algorithms cause affective polarization, i.e., hostility toward opposing political parties (*19*–*22*)? This concern has grown since the 2016 US presidential election (*23*), and the debate remains ongoing after the 2020 and 2024 elections. If social media algorithms are causing affective polarization, they might not only bear responsibility for rising political incivility online (*24*), they might also pose a risk to trust in democratic institutions (*25*). In this case, isolating the algorithmic design choices that cause polarization could offer alternative algorithmic approaches (*26*).

A major hypothesized mechanism for how feed algorithms cause polarization is a self-reinforcing engagement loop: users engage with content aligning with their political views, the feed algorithm interprets this engagement as a positive signal, and the algorithm exposes even more politically aligned content to users, leading to a polarizing cycle. Some studies support this hypothesis, finding that online interactions exacerbate polarization (*27*), potentially because of the increased visibility of hostile political discussions (*28*), divisive language (*29*–*33*), and content that reinforces existing beliefs (*34*). However, large-scale field experiments aimed at reducing polarization by intervening on the feed algorithm—for example, by increasing exposure to out-party content—have found both a decrease (*35*) and an increase (*36*) in polarization. Similarly, recent large-scale experiments on Facebook and Instagram found no evidence that reduced exposure to in-party sources or a simpler reverse-chronological algorithm affected polarization and political attitudes (*23*, *37*) during the 2020 US election. These mixed results reveal the difficulty in identifying what, if any, algorithmic intervention might help reduce polarization, especially during politically charged times.

We distilled the goals of these prior interventions to a direct hypothesis that we could operationalize through real-time LLM reranking: that feed algorithms cause affective polarization by exposing users specifically to content that polarizes. An algorithm that up-ranks content reflecting genuine political dialogue is less likely to polarize than one that up-ranks demagoguery. This content-focused hypothesis has been difficult to operationalize into interventions, making studies that intervene on cross-partisan exposure and reverse-chronological ranking attractive but more diffuse in their impact and thus more likely to observe mixed results. However, by connecting our real-time reranking infrastructure with recent advances in LLMs, we could create a ranking intervention that more directly targets the focal hypothesis (*38*) without needing platform collaboration. We drew, in particular, on a recent large-scale field experiment that articulated eight categories of antidemocratic attitudes and partisan animosity as bipartisan threats to the healthy functioning of democracy (*39*). We operationalized these eight categories into an artificial intelligence (AI) classifier that labels expressions of these constructs in social media posts, does so with accuracy comparable to trained annotators, and produces depolarization effects in a lab setting on a fixed feed (*40*). This real-time classification enabled us to perform a scalable, content-based reranking experiment on participants' own feeds in the field (*41*).

We conducted a preregistered field experiment on X, the most used social media platform for political discourse in the US (*42*), using our extension to dynamically rerank participants' social media content by either increasing or decreasing exposure to content that expresses these eight factors of antidemocratic attitudes and partisan animosity (AAPA) over the course of a week. The experiment was conducted during a pivotal moment in the 2024 US election cycle, from July to August 2024, an important period for understanding how social media feeds impact affective polarization. Major political events during the study period included the attempted assassination of Donald Trump, the withdrawal of Joe Biden from the 2024 presidential race, and the nomination of Kamala Harris as the Democratic Party's candidate. These events allow us to examine the impact of heterogeneous AAPA content on partisan polarization and hostility. We measured the intervention's effect on affective polarization (*43*) and emotional experience

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(*44*). Compared with control conditions that did not rerank the feed, decreased AAPA exposure led to warmer feelings toward the political outgroup, whereas increased AAPA exposure led to colder feelings. These changes also affected participants' levels of negative emotions (anger and sadness) as measured through in-feed surveys.

Exposure to antidemocratic attitudes and partisan animosity

There are many viable theories as to which content, if amplified, could impact affective polarization. In this experiment, we drew specifically on prior work that articulates eight factors as problematic outcomes for healthy democratic functioning, referred to as AAPA (*39*): (i) partisan animosity, (ii) support for undemocratic practices, (iii) support for partisan violence, (iv) support for undemocratic candidates, (v) opposition to bipartisanship, (vi) social distrust, (vii) social distance, and (viii) biased evaluation of politicized facts [supplementary materials (SM) section S3.1]. We hypothesized that increased or decreased exposure to AAPA content causes corresponding changes in affective polarization. Our feed-ranking intervention first identified only political content in the feed using a broad definition from the Pew Research Center (*45*) that includes mentions of officials and activists, social issues, news, and current events (SM section S4.3). We then scored each political post from zero to eight based on the number of the eight AAPA factors expressed in its text, and we classified a post as AAPA if it reflected at least four of these eight factors. This scoring was performed by a large language model (SM section S4.4), which showed a high correlation with human judgment on identifying these eight factors in social media posts (*40*).

Experimental design

We designed two parallel experiments to measure the impact of reducing and increasing exposure to AAPA posts (Fig. 1). The first, a "Reduced Exposure" experiment, down-ranked all AAPA content, pushing it lower in the participants' feeds. By contrast, the second, an "Increased Exposure" experiment, up-ranked AAPA posts in their feeds. These posts came from the inventory of posts recommended by X to all participants with the same political leaning. Both experiments were randomized controlled trials. After providing informed consent and installing our web extension, participants were randomly assigned to one of the two experiments and then randomly divided into treatment or control conditions for each experiment. The overall experiment spanned 10 days, with the first 3 days serving as a baseline period, during which no intervention was applied. During the subsequent 7 days, participants in the treatment groups received the modified feed (with down-ranked or up-ranked AAPA posts), whereas those in the control groups continued to receive the original, unmodified feed.

We assessed outcomes at two different time points: post-experiment and in-feed. To measure post-experiment effects, we administered a survey asking participants how warmly they feel toward members of the opposing party (0 to 100 "degrees" on a feeling thermometer) and about the various emotions (e.g., anger, sadness, excitement, or calm) they felt when reading social media over the previous week (five-point scale between "never" and "all the time"). We surveyed the participants both before and after the experiment, enabling us to quantify the impact of the intervention at the end of the week-long treatment (SM section S1.2).

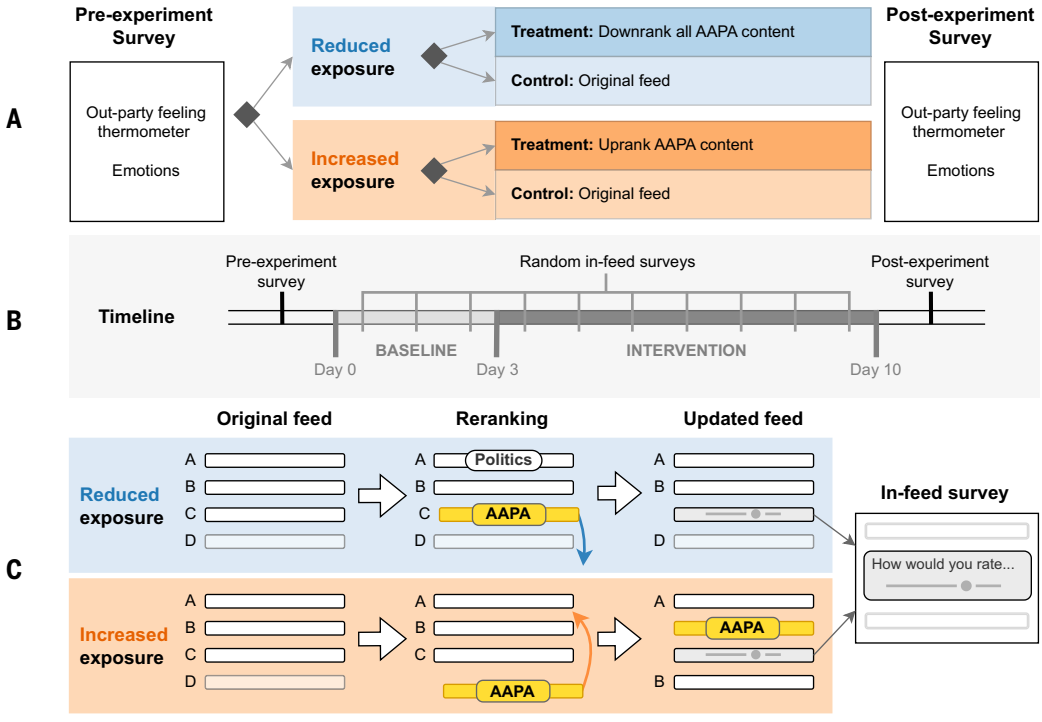


Fig. 1. Overview of the field experiment on X. (**A**) Participants completed a pre-experiment survey to rate their feelings toward the opposing party and report their emotions. Then, they were randomly assigned to one of the two parallel experiments, reduced or increased AAPA content exposure, and further randomized into treatment and control groups. After 10 days, they completed a post-experiment survey. (**B**) Timeline of the experiment. During the first 3 days, no intervention was applied to measure the participants' baseline responses. For the next 7 days, participants in the treatment groups received increased or decreased AAPA exposure. In-feed surveys were periodically shown throughout the baseline and the intervention period. (**C**) The interventions down- or up-ranked AAPA posts in the participants' feeds depending on their treatment assignment.

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To measure in-feed effects, we added posts containing a small survey directly in the participant's feed. These surveys, modeled on ecological momentary assessments (*46*), resembled typical social media posts but were colored to stand out and contained a slider survey instrument. Each survey randomly selected one of two outcomes to assess: (i) affective polarization, where participants rated their feelings toward the opposing party, or (ii) emotional experience, where participants reported the extent to which they felt two emotions, one randomly selected positive emotion (excited or calm) and one randomly selected negative emotion (angry or sad). For both experiments, the in-feed surveys were shown at an equivalent rate and position in the control conditions (SM section S1.2). To reduce fatigue effects, each time participants completed the in-feed survey, they were less likely to receive another one on the same day.

Feed reranking

To administer the interventions, we developed a browser extension for Google Chrome and Microsoft Edge that reranked participants' feeds in real time (*41*). The extension activated when participants accessed the "For you" feed on X. It intercepted the participants' original feed and sent it to a remote backend, which scored and reranked the posts (SM section S4). This process added a latency of about three seconds when opening the feed for the first time, which was consistent across all experimental conditions. To rerank the original feeds, we first identified political posts (SM section S4.3), then classified whether and how much AAPA the political posts contained.

In the Reduced Exposure treatment condition, we down-ranked all AAPA posts in the participants' feeds. If participants continued to scroll down far enough, e.g., across several incremental loads of more content, these posts would appear (fig. S12). Before downranking all the AAPA posts, we randomly selected the position of one AAPA post and, if the in-feed survey was sampled to be included, inserted the post with the survey in the next available position (SM section S3.2.1). This survey position was consistent across treatment and control conditions. Thus, participants in the control condition saw AAPA posts and the in-feed survey, whereas participants in the treatment condition saw the survey in the position where the first AAPA post used to be, but all AAPA posts were placed further down. The survey was limited to only a few times per day to avoid participant fatigue (SM section S1.2). In the Increased Exposure experiment, because we lacked access to the platform's full post inventory, we sourced AAPA posts from the inventory of all posts in the "For you" feeds of other study participants with the same political leaning. We selected a random position in the original feed and up-ranked a random AAPA post to that position. The post containing the survey, if included, was inserted in the next available position. In the control condition, the survey was still added in the randomly sampled position, but no post was up-ranked. As in the Reduced Exposure experiment, the survey was placed consistently in the treatment and control conditions. During continuous scrolling sessions, this process repeated approximately every 30 posts (SM section S3.2.2).

The Reduced Exposure treatment down-ranked 85.4 AAPA posts on average (median = 22) per user per day, and the Increased Exposure up-ranked 10.9 AAPA posts on average (median = 5.43) per user per day (SM section S3.2.3). To avoid excessive exposure to polarizing content, we up-ranked fewer but higher-scoring AAPA posts. Despite the difference in the number of posts affected by the two interventions, the average AAPA score of the feed consumed during treatment days exhibited symmetric changes across the two experiments (fig. S11).

Results

We recruited participants from two online survey platforms: CloudResearch and Bovitz (*47*). We targeted participants who used X, were age 18 or older, resided in the United States, and self-identified as either Democrats or Republicans (SM section S5). After giving informed consent, participants installed our web extension. To ensure that the participants had enough political content, we used a prescreening task to filter for users whose "For you" feeds contained at least 5% of posts related to politics or social issues, retaining 73.4% of the screened participants. The participants were recruited during the US election season, between 7 July and 14 August 2024, and instructed to use X daily only from the web interface.

We preregistered a target of at least 1100 participants to complete the full study (600 in the Reduced Exposure experiment and 500 in the Increased Exposure experiment) based on power analyses performed on data from a pilot experiment (*48*). Toward this goal, a total of 1662 participants completed the initial survey and qualified for inclusion in the study. Participants were randomly assigned to either the Reduced or Increased Exposure experiment until the preregistered experiment-specific quotas were reached. Of the qualified participants, 1256 completed the full study by responding to a follow-up survey at the end of the experiment, 727 of which were in the Reduced Exposure experiment and 529 of which were in the Increased Exposure experiment (fig. S1 shows the participant recruitment funnel). Both treatment and control groups were well balanced across covariates, and there was no evidence of differential attrition between conditions (SM section S1.3). To maintain a consistent sample across the post-experiment and in-feed analyses, the reported treatment effects are based on participants who completed both the post-experiment survey and at least one in-feed survey for the corresponding outcome (N = 1090). The results are similar when the full population is considered (SM sections S11.1 and S11.2).

Participants received $20 upon completing the study. The sample included 48% women, 50% men, and 2% who identified as another gender. Participants ranged in age from 18 to 81 years (mean = 41.6, SD = 14.0). Party affiliation was 66% Democrat and 34% Republican. Educational attainment was as follows: 17 (1%) had less than a high school degree, 158 (13%) were high school graduates, 267 (21%) had some college but no degree, 145 (12%) held an associate degree, 419 (33%) held a bachelor's degree, 189 (15%) held a master's degree, 30 (2%) held a professional degree (e.g., JD, MD), and 31 (2%) held a doctoral degree. Participants reported their race or ethnicity and were allowed to choose multiple responses. Overall, 945 (75%) identified as white, 183 (15%) identified as Black or African American, 153 (12%) identified as Hispanic or Latino, 66 (5%) identified as East Asian, 34 (3%) identified as South Asian, 28 (2%) identified as American Indian or Alaska Native, 5 (<1%) identified as Middle Eastern, and 9 (<1%) identified as other.

On average, participants viewed 154 posts per day (median = 92) and spent 24.1 min (median = 13.1 min) per day with a browser tab in the foreground on their X feed. During the baseline period, an average of 31.6% of the posts that participants viewed were related to politics or social issues. Of these posts, 32% were classified as AAPA, which comprised an average of 10.1% of the total feed content.

The interventions had the intended effects (fig. S9). During the baseline period, participants across all experimental conditions were exposed to a similar fraction of AAPA posts. However, after the intervention began, participants in the treatment condition of the Reduced Exposure experiment experienced a substantial decrease in exposure to such posts (treatment, 1.04% AAPA posts of all feed exposures; control, 10% AAPA posts of all feed exposures; 89.6% relative reduction), whereas those in the treatment condition of the Increased Exposure experiment were exposed to more AAPA posts (treatment, 13.4% AAPA posts of all feed exposures; control, 10.6% AAPA posts of all feed exposures; 26.4% relative increase).

Over the full 10 days of the experiment, participants answered an average of 13 (median = 9) in-feed affective polarization questions in the Reduced Exposure experiment and 14.7 (median = 11) in the Increased Exposure experiment. Similarly, they answered an average of 13.2 (median = 9) in-feed emotion surveys in the Reduced Exposure

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experiment and 15.0 (median = 12) in the Increased Exposure experiment, with each survey asking them to rate two emotions.

In the post-experiment survey, we asked participants whether the browser extension had impacted their experience on X. Across conditions, most participants reported that they did not notice any impact on their online experience (74.2%), whereas others guessed changes that were not part of the intervention (SM section S11.4). These results indicate a limited awareness of the experimental feed-ranking intervention, reducing the likelihood that the observed results were due to demand effects.

Affective polarization

Exposure to AAPA significantly influenced affective polarization (Fig. 2). Participants in the reduced AAPA exposure experiment reported a significant post-experiment effect corresponding to an out-party warmth increase of 2.11 degrees [95% confidence interval (CI): 0.15, 4.06; $P = 0.035$]. Conversely, participants in the Increased Exposure experiment exhibited a symmetric reduction in affective warmth toward the opposing party of −2.48 degrees (95% CI: −4.79, −0.17; $P = 0.036$). We did not observe any significant heterogeneity in treatment effects across the preregistered moderators (SM section S8), including party identification, indicating that the effect was consistent and bipartisan. Similar to other studies (*49*), we contextualize these effects against the rising levels of partisan animosity in American society over time: Our effects correspond to a decrease and increase in affective polarization equivalent to 3 and 3.6 years, respectively. We observed even stronger effects with the in-feed assessment, with Reduced Exposure causing an out-group warmth increase of 3.24 degrees (95% CI: 1.21, 5.27; $P = 0.002$) and Increased Exposure causing a cooling of −2.56 degrees (95% CI: 4.53, −0.59; $P = 0.011$). Supplementary exploratory analysis reweighting our sample to match the population of X users reduced our statistical power but replicated our results, indicating substantial robustness (SM section S10). The only exception was the posttreatment effects in the Reduced Exposure experiment, which had a similar point estimate but were not statistically significant.

In an exploratory analysis, we investigated which types of AAPA content were predictive of changes in affective polarization. Specifically, a higher proportion of posts supporting partisan animosity (95% CI: −13.49, −2.61; $P = 0.003$), opposition to bipartisan cooperation (95% CI: −26.55, −4.40; $P = 0.006$), and biased evaluation of politicized facts (95% CI:−12.83, −2.21; $P = 0.005$) predicted a colder feeling toward the opposing party by the end of the study (fig. S22).

In addition to affective polarization (or partisan animosity), we also surveyed participants about their attitudes toward the remaining seven AAPA factors. We found that, although reduced and increased exposure to AAPA content impacted affective polarization, it had no significant effects on these other outcomes (SM section S7). These results are consistent with recent studies finding that changes in affective polarization are not necessarily linked to changes in antidemocratic attitudes (*39*, *50*).

Emotions

Changes to the feed algorithm caused in-feed but not post-experiment changes to participants' negative emotions (Fig. 3). Reduced Exposure caused a change of −5.05 points [95% CI: −7.68, −2.42; adjusted P (P_{adj}) = 0.002] for "angry" and −3.68 points (95% CI: −6.42, −0.94; $P_{adj} = 0.024$) for "sad." Likewise, in the Increased Exposure, participants reported a symmetric increase in negative emotions: 5.13 points (95% CI: 2.05, 8.22; $P_{adj} = 0.013$) for "angry" and 4.38 points (95% CI: 1.00, 7.76; $P_{adj} = 0.036$) for "sad." Neither intervention had an effect on the positive emotions "calm" and "excited." Post-experiment emotion effects were not significant, suggesting that the effects were likely short-lived.

Engagement

We tested whether our interventions impacted platform engagement (SM section S9). Neither experiment observed differential attrition rates across conditions.

In the Reduced Exposure experiment, participants in the treatment group had a similar number of daily sessions (estimate, −0.06; 95% CI: −0.16, 0.04) but spent −4.71 fewer minutes per day (95% CI: −8.22, −1.20) compared with that of participants in the control group. Over the 7-day intervention period, they viewed −169.52 fewer posts (95% CI: −325.48, −13.56), liked −24.06 fewer posts (95% CI: −39.53, −8.59), but reposted a similar number of posts (estimate, −4.18; 95% CI: −9.89, 1.53). On the other hand, they engaged with the posts they viewed at higher rates, with a 0.61% higher like rate (95% CI: 0.11%, 1.12%) and a 0.058% higher repost rate (95% CI: 0.009%, 0.11%). Although the per-view engagement rates increased, these findings suggest that down-ranking AAPA content comes at the cost of slightly reduced engagement overall.

In the Increased Exposure experiment, we did not find any significant differences in engagement between the treatment and control groups. Participants in the treatment group had a similar number of daily sessions (estimate, −0.04; 95% CI: −0.16, 0.08) and spent a comparable amount of time on the platform (estimate, −1.76 min per day; 95% CI: −6.17, 2.66). They also viewed (estimate, −132.3; 95% CI: −292.5, 27.94), liked (estimate, 10.25; 95% CI: −5.23, 25.74), and reposted (estimate, 3.38; 95% CI: −0.06, 6.82) a similar number of posts and liked and reposted the posts they viewed at similar rates (effect on like rate, 0.22%; 95% CI: −0.57%, 1.01%; effect on repost rate, −0.052%; 95% CI: −0.17%, 0.066%).

Discussion

This study demonstrates a deployable framework for conducting re-ranking field experiments on social media feeds without requiring direct involvement from the platforms. We developed a browser extension infrastructure that enables real-time manipulation of what consenting users see in their feed, using LLM-based classifiers to dynamically assess and rerank posts based on a custom ranking function. This approach opens new avenues for auditing, evaluating, and prototyping alternative feed-ranking strategies in real-world settings.

Our specific experimental findings demonstrate that repeated exposure to AAPA posts, particularly during heightened political tensions such as the months leading up to the 2024 US election, significantly influences users' affective polarization and emotional responses. A feed-ranking algorithm that controls exposure to such content produces measurable changes, equivalent to ~3 years of affective polarization, over the period it was deployed. Notably, our results indicate that the intervention does not disproportionately impact any specific

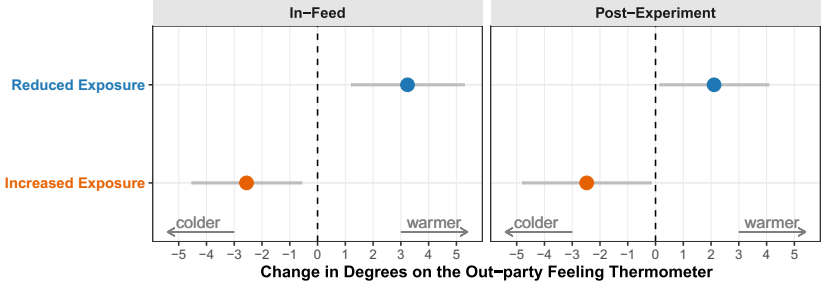


Fig. 2. Effects of reducing and increasing exposure to AAPA content in participants' feeds on their feeling toward the out-party relative to the corresponding control group. (Left) Participants were surveyed within the feed during the intervention and (right) off-platform after the experiment. The feeling thermometer scale was between 0 (cold) and 100 (warm). The error bars represent 95% CIs.

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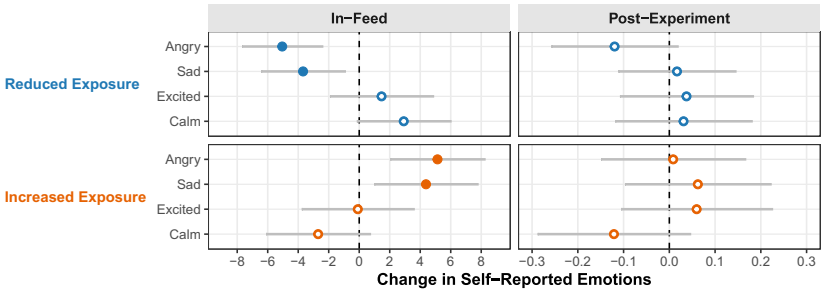


Fig. 3. Effects of reducing and increasing exposure to AAPA content in participants' feeds on their experiences of emotion compared with that of the corresponding control group. (Left) Participants were surveyed within the feed during the intervention [scale ranged from 0 ("none at all") to 100 ("extremely")] and (right) off-platform after the experiment [scale ranged from 1 ("never") to 5 ("all the time")]. The filled circles represent statistical significance ($P_{adj} < 0.05$, adjusted for multiple hypothesis testing), and the error bars represent 95% CIs.

political leaning or demographic group, suggesting that the approach is bipartisan. The success of this method shows that it can be integrated into social media AI to mitigate harmful personal and societal consequences. At the same time, our engagement analyses indicate a practical trade-off: Interventions that down-rank AAPA may reduce short-term engagement volume, posing challenges for engagement-driven business models and supporting the hypothesis that content that provokes strong reactions generates more engagement.

Methodological advances and policy implications

Our work opens the door to conducting large-scale causal inference on social media feed algorithms without requiring "independence by permission" (*51*) from platforms. Although prior work made important advances through randomized controlled trials on the Facebook and Instagram algorithms (*23*, *37*), these experiments could only proceed with negotiated support from Meta's executives, an arrangement that may limit the scope and independence of scientific work. Our approach enables scientists to test a much broader set of interventions beyond those that platforms might agree to test, fostering a more robust ecosystem for independent social media research.

Limitations and future directions

Our study focused on a sample of partisan US users restricted to the web version of X during the months leading up to a US presidential election. This context, marked by a high volume of political content and emotional engagement, enabled us to test interventions effectively but may limit the generalizability of the findings beyond a period of heightened political activity (*52*, *53*). Our population was limited to X users with at least 5% of political content, but we intentionally selected this group to ensure that the interventions were viable.

Additionally, the experiment examines short-term effects—immediate in-feed responses and post-experiment survey responses measured soon after the experiment—but does not address whether these effects persist over time. In additional analysis examining the variation of the treatment effects by experiment day and the time gap between the end of the experiment and when the participants started the post-experiment survey, we did not find any evidence for immediate decay of the observed effects (SM section S11.5). We also note that in scenarios where this ranking approach is integrated into the feed algorithm, the intervention would be continuously applied, and any effects may dissipate once it is turned off. Other technical limitations of our approach include the difficulty in performing cluster-randomized experiments testing for social influence (*54*) and the inability to collect the full feed inventory for a given user.

Our reliance on in-feed surveys to assess immediate responses to stimuli follows established practices of ecological momentary assessment (*46*). To mitigate the risk of demand effects associated with the close proximity of the assessment, we followed best practices in designing the experiment (*55*), providing only essential information about the study's purpose. Post-experiment survey responses to an open-ended question about participants' experiences during the study suggest that these precautions were effective, as they did not report any impact on their experience attributable to their assigned condition (SM section S11.4).

Describing the causal relation between social media and polarization as a unified phenomenon remains challenging: Platforms are complex sociotechnical systems, and small variations in design or exogenous factors can lead to different outcomes (*56*). Future longitudinal research should investigate the impact of sustaining the intervention over longer periods of time—especially on long-term metrics, such as user retention (*57*) and news knowledge—and evaluate the efficacy of feed reranking across different platforms and cultural settings. For example, open questions remain about how our approach can be extended to the broader online information ecosystem and whether our findings hold true in different political systems with specific cultural contexts or party structures. In addition, future research can move beyond the initial eight factors related to antidemocratic attitudes, operationalize new factors grounded in other social science theories (*38*), and measure how this content can impact other experiences online, such as mental health, trust in institutions, civic engagement, or the quality of democratic discourse. By using a design similar to that of our study, these future investigations can also examine different time horizons, from immediate reactions captured through in-feed surveys to long-term impacts assessed through follow-up surveys conducted days or even months later.

As political polarization and societal division become increasingly linked to social media activity, our findings provide a potential pathway for platforms to address these challenges through algorithmic interventions. Together, these interventions may result in algorithms that not only reduce partisan animosity but also promote greater social trust and healthier democratic discourse across party lines (*38*, *40*).

Materials and methods summary

Participant recruitment

We recruited participants through two online platforms, CloudResearch and Bovitz, targeting US residents over 18 years old who self-identified as either Republican or Democrat and were active users of X (SM section S1.1). Qualified individuals were invited to complete a screening task, which included installing a browser extension that analyzed their X feed. To ensure the interventions could have a meaningful impact on participants' feeds, only those with at least 5% of posts related to politics or social issues were invited to participate. Figure S1 summarizes the recruitment funnel, including the number of individuals at each stage of the process. Participants were not instructed to use X in any particular way, but they received daily reminders if they had not used the platform that day.

Surveys

Participants were required to complete a pre-experiment survey that included questions about their demographics, personal values, affective experiences on social media, and political attitudes (SM section S1.2). At the end of the study, they were asked to complete a post-experiment survey with a similar structure but without questions on demographics or personal values. The surveys took about 20 and 10 min to complete, respectively. The section of the survey measuring political attitudes included a question related to our primary outcome, assessing participants' feelings toward out-partisans, while the section on affective

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experiences on social media included questions related to our secondary outcomes, measuring how often participants felt angry, sad, excited, and calm.

Throughout the experiment, we also inserted short surveys into participants' feeds to measure their feelings in the moment. Each in-feed survey included either one question about participants' feelings toward members of the out-party or two questions asking how much they felt one positive emotion (excited or calm) and one negative emotion (angry or sad), randomly selected. To minimize survey fatigue, once a participant answered a question, they were not shown the same type of question for the next 10 minutes and were subsequently less likely to receive that question again.

Feed reranking implementation

To administer the interventions, we developed a browser extension that intercepted the participants' original X feeds, sent the posts to a remote backend for real-time analysis, and displayed the reranked feeds (SM section S3.2). The backend used a fast, custom-built classifier to detect political posts (SM section S4.3), and OpenAI's GPT-3.5 Turbo model to score the political posts on the eight AAPA dimensions (SM section S4.4). The reranking process took approximately 3 seconds and was consistent across all experimental conditions. The extension also administered in-feed surveys and logged participants' engagement with the feed, including which posts they viewed, liked, and reposted.

Experimental design

We used Bernoulli randomization to assign participants to either the Reduced or Increased Exposure experiment, and subsequently to treatment and control groups with equal probability (SM section S1.3). Owing to differences in how the two interventions were operationalized, each experiment had its own control group. To determine the sample size, we conducted a power analysis based on data from a pilot experiment with 70 participants that followed the same experimental design and analysis procedure. We chose target quotas that enabled us to detect effects of at least two degrees on the feeling thermometer scale for affective polarization with 95% power. We used randomization inference to test for imbalances in participants' characteristics, differential attrition rates, and differential attrition patterns between the treatment and control groups in each experiment. We found no evidence of covariate imbalance or differential attrition.

Analysis

As preregistered, to estimate the effects of the interventions on the post-experiment outcomes, we used ordinary least squares, regressing the outcome variable on the treatment assignment indicator, the participant's response to the same question on the pre-experiment survey, and a recruitment platform indicator (SM section S1.4). To estimate the effects on the outcomes measured with in-feed surveys, we used linear mixed-effects models to account for the multiple observations per participant. We regressed the outcome on the treatment assignment, the participant's average response on the same question during the 3-day baseline period, and the recruitment platform indicator. To estimate heterogeneous treatment effects, we used the same model specifications but included an interaction between the treatment assignment and the moderator, running one regression per moderator (SM section S8). As preregistered, we considered party identification, party strength, and the proportion of political content in the participant's feed during the baseline period as primary moderators, and gender, race, age, income, and socioeconomic ladder as secondary moderators. To account for testing multiple hypotheses, we applied false discovery rate adjustment (SM section S6). Finally, to appropriately model the nature of the outcomes, we used quasi-Poisson regression to estimate the effects on engagement volume and Beta regression to estimate the effects on engagement rates (SM section S9).

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ACKNOWLEDGMENTS

We thank P. Barbera, D. Eckles, M. Gentzkow, D. Ong, J. Pan, S. Pentland, J. Ugander, R. Willer, and the anonymous reviewers for helpful feedback and discussions. M.B. discloses an additional affiliation as a cofounder of Simile AI, Inc., which builds AI human behavioral simulations. **Funding:** This work was supported in part by the National Science Foundation under awards IIS-2403433, IIS-2403434, IIS-2403435, and BCS-2214203; the Swiss National Science Foundation under award P500PT-206953; and a Hoffman-Yee grant from the Stanford Institute for Human-Centered Artificial Intelligence. **Author contributions:** Conceptualization: T.P., M.S., C.J., J.T., J.H., M.B.; Methodology: T.P., M.S., C.J., J.T., J.H., M.B.; Software: T.P.; Validation: T.P., M.S.; Formal analysis: T.P., M.S.; Investigation: T.P., M.S.; Resources: T.P., M.S., C.J.; Data curation: T.P., M.S.; Writing – original draft: T.P., M.S., C.J., M.B.; Writing – review & editing: T.P., M.S., C.J., J.H., J.T., M.B.; Supervision: J.T., J.H., M.B.; Project administration: T.P., M.S., C.J., M.B.; Funding acquisition: M.S., C.J., J.H., M.B. **Competing interests:** The authors declare that they have no competing interests. **Data and materials availability:** The repository associated with the manuscript, containing aggregated data and code required to reproduce the paper results, is available at Dryad (*58*). The preregistration of the study is available at OSF (*48*). **License information:** Copyright © 2025 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. https://www.science.org/about/science-licenses-journal-article-reuse

SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.adu5584
Methods and Materials; Ethical Considerations; Supplementary Text; Figs. S1 to S24; Tables S1 to S32; References (*59*–*81*); MDAR Reproducibility Checklist
Submitted 21 November 2024; resubmitted 14 August 2025; accepted 29 September 2025
10.1126/science.adu5584

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